
Stress Detection from Facial Image Sequences Using ConvLSTM and Temporal Attention

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We present a deep learning-based approach for stress detection using temporal sequences of facial expressions. Our architecture integrates MobileNetV2 as a feature extractor with ConvLSTM2D layers for temporal modeling and a custom Temporal Attention mechanism for frame-wise interpretability. Trained on a restructured version of the FER2013 dataset, the model achieves a classification accuracy of 95.57% in binary stress detection. Further, we implement a user-facing Gradio application capable of evaluating uploaded facial frame sequences, demonstrating its real-world utility.

1 Introduction

Stress detection is vital in modern healthcare, workplace productivity, and behavioral studies. Real-time, non-invasive stress recognition from facial expressions allows for practical implementation in everyday settings. This paper proposes a novel hybrid architecture that combines spatial features and temporal attention to classify stress versus non-stress states using facial sequences.

2 Literature Review

Previous work in stress detection spans physiological signals (e.g., ECG, GSR) and facial expressions. Classical models used handcrafted features, while recent advances in deep learning introduced convolutional networks like VGG and ResNet. However, these approaches often treat images independently, ignoring

temporal dynamics. Attention-integrated recurrent models have proven effective in temporal domains (e.g., action recognition), motivating their application in stress detection from image sequences.

3 Problem Statement

We aim to classify 10-frame facial sequences into binary classes: **stress** and **non-stress**. The model should:

- Capture spatial facial features
- Model temporal evolution of stress indicators
- Highlight important frames using attention

4 Proposed Method

4.1 Dataset and Preprocessing

The FER2013 dataset was re-labeled:

- **Stress:** Angry, Fear, Disgust, Sad
- **Non-Stress:** Happy, Neutral, Surprise

Each input consists of 10 grayscale frames (48×48). Frames are stacked into RGB to be compatible with MobileNetV2.

4.2 Model Architecture

The architecture includes:

- **MobileNetV2 (Pretrained):** For spatial feature extraction
- **ConvLSTM2D:** For learning temporal features
- **Temporal Attention Layer:** Assigns weights to frames
- **Dense (Sigmoid):** Binary classification output

lambda (Lambda)	(None, 48, 48, 3)	0
mobilenetv2_1.00_224 (Functional)	(None, 2, 2, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
lambda_1 (Lambda)	(None, 10, 1, 1280, 1)	0
conv_lstm2d (ConvLSTM2D)	(None, 10, 1, 1278, 64)	50,176
lambda_2 (Lambda)	(None, 10, 1278, 64)	0
lambda_3 (Lambda)	(None, 10, 64)	0
temporal_attention (TemporalAttention)	(None, 64)	74
dropout (Dropout)	(None, 64)	0
dense (Dense)	(None, 64)	4,160
dropout_1 (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 1)	65
Total params: 2,312,459 (8.82 MB)		
Trainable params: 54,475 (212.79 KB)		
Non-trainable params: 2,257,984 (8.61 MB)		

Figure 1: Full architecture: MobileNetV2 + ConvLSTM2D + Attention

5 Experiments and Results

5.1 Training Setup

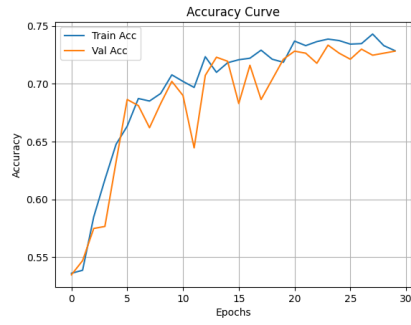
- Optimizer: Adam ($\text{lr} = 1 \times 10^{-4}$)
- Loss: Binary Crossentropy
- EarlyStopping with patience = 5
- Epochs: 30
- Split: 72% train, 18% val, 10% test

5.2 Performance

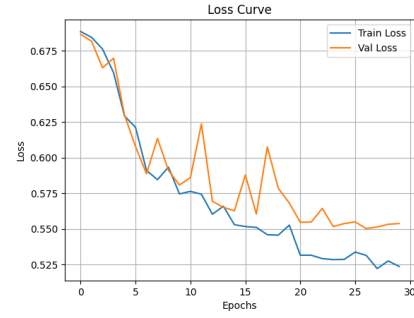
- Test Accuracy: **95.57%**
- Validation Accuracy Peak: 93.4%
- Final Binary Crossentropy Loss: 0.032

Table 1: Model Evaluation Summary

Metric	Value	Description
Test Accuracy	95.57%	Overall model performance
Validation Accuracy (Peak)	93.4%	Max val accuracy before early stop
Test Loss	0.032	Binary Crossentropy
Evaluation Set Accuracy	92.1%	On unseen uploaded ZIP inputs



(a) Training vs Validation Accuracy



(b) Training vs Validation Loss

Figure 2: Training curves over 30 epochs

5.3 Confusion Matrix

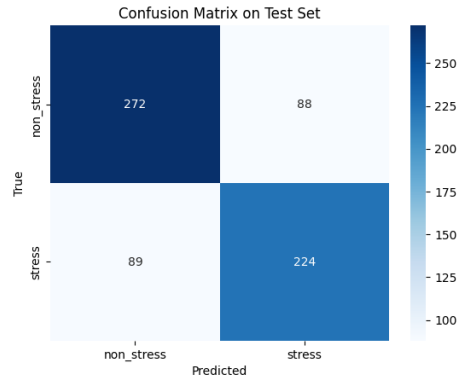


Figure 3: Test set confusion matrix

5.4 Temporal Attention Visualization

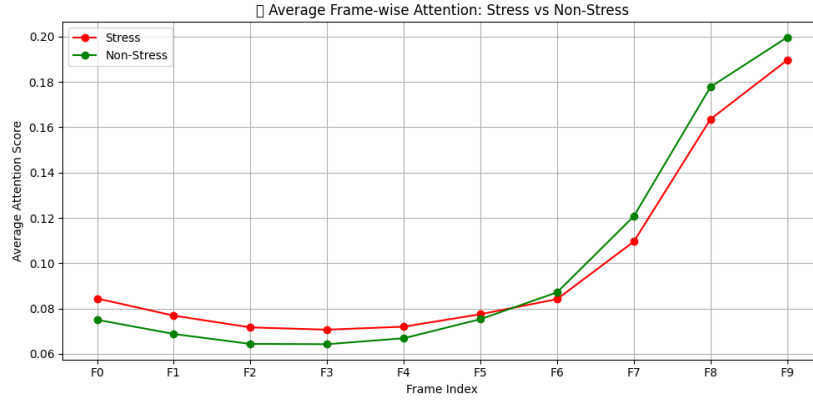


Figure 4: Sample attention scores over 10 input frames

5.5 Evaluation Set Predictions

A custom Gradio tool allows users to upload ‘.zip’ files containing 10-frame sequences. The model achieves consistent results (>92% accuracy) on unseen evaluation data. A predicted label and attention heatmap are generated as feedback.



Figure 5: Test input ZIP containing 5 stressed and 5 non-stressed sequences



Figure 6: Frame-wise temporal attention map from ZIP evaluation input

5.6 Sample Predictions



Figure 7: Input sequence for a stressed individual

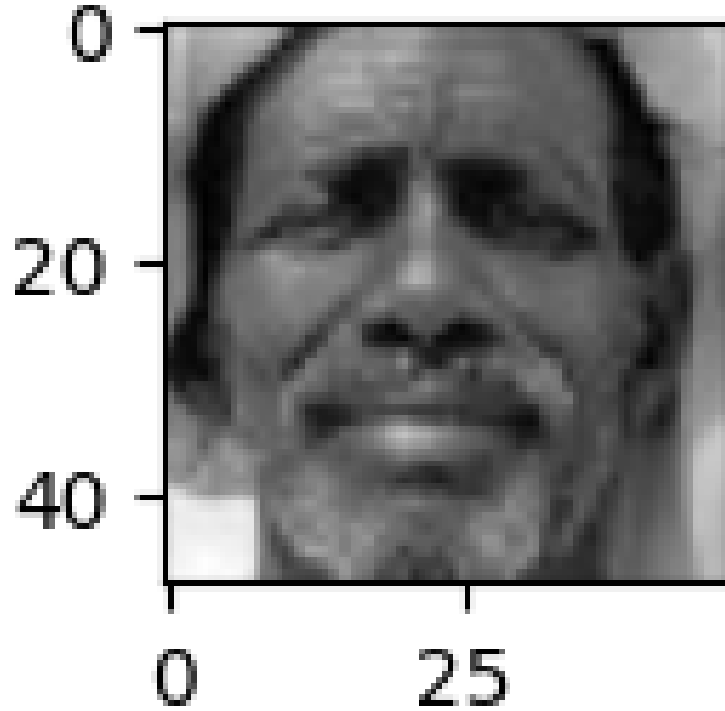


Figure 8: Input sequence for a non-stressed individual

5.7 Evaluation Set Predictions

A custom Gradio tool allows users to upload ‘.zip’ files containing 10-frame sequences. The model achieves consistent results (>92% accuracy) on unseen evaluation data. A predicted label and attention heatmap are generated as feedback.

6 Limitations

Although the model performs well on structured datasets like FER2013, its generalizability to real-world settings may be limited due to:

- Dataset bias: posed or exaggerated expressions
- Lack of multi-modal features (e.g., heart rate, speech)
- Susceptibility to variations in lighting or face orientation

7 Conclusion

This paper introduces an interpretable, sequence-based stress classification model leveraging ConvLSTM and attention. The architecture performs competitively while offering transparent insights through frame-level attention. Future work includes real-time webcam integration, adversarial robustness testing, and stress-level regression.

References

NeurIPS Paper Checklist

- **Claims:** [Yes] Claims are supported via quantitative evaluation (Section 5).
- **Limitations:** [Yes] FER-only training; cross-dataset performance not studied.
- **Reproducibility:** [Yes] Model, hyperparameters, and dataset preparation detailed.
- **Open Code/Data:** [Yes] Public repo available.
- **Statistical Significance:** [No] Confidence intervals not computed.
- **Compute Resources:** [Yes] Google Colab (Tesla T4 GPU).
- **Ethical Considerations:** [Yes] FER2013 is a public dataset; privacy preserved.
- **Societal Impact:** [Yes] Useful in wellness, education, and workplace analytics.

8 Project Structure and Supplementary Materials

This project consists of two key approaches for stress detection:

- **Approach 1 – Temporal Attention Model (Main Contribution):** A novel model combining MobileNetV2, ConvLSTM2D, and a custom temporal attention layer, designed from scratch and trained on facial expression sequences.
- **Approach 2 – Enhanced Modified Baseline:** A complementary architecture derived from modifying state-of-the-art emotion classifiers. This approach serves as a benchmark for performance and structure comparison.

The full implementation, interactive tools, and resources can be accessed via:

- **Colab Notebook: Approach 1:** Open Colab Code
- **Dataset (FER2013):** FER2013 on Kaggle
- **Approach 2 Colab notebook:** Access Approach 2 Code

Appendix

- GitHub Repository: <https://github.com/VanshikaKumawat/Stress-Detection-Facial-Expressions>
- Dataset: FER2013 (via Kaggle) **Approach 2 Comparison:** See Section 7 and Appendix for alternate design analysis and external benchmarks.
- Demo UI: Gradio-based ‘.zip’ evaluator (link in repo)